Bridging Neural and Behavioural Data with XAI in Predictive Neuromarketing

Final Year Research Project

Sri Lanka Institute of Information Technology

Department of Computer Science

Project Code: 25-26J-145-DS

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Problem & Impact

Problem

- Behavioral data like gaze and reaction time miss deeper neural processes
- Existing neuromarketing models rarely combine neural and behavioral data
- Limited explainability in AI predictions
- Ethical risks such as bias and manipulation

Impact

- Poor understanding of consumer behavior
- Ineffective marketing strategies
- Lack of trust in AI-driven insights

> *"Current approaches fail to provide a holistic view of consumer decision-making processes"*

Proposed Solution

Integrated Framework

Combining EEG neural signals and behavioral metrics

Reinforcement Learning

Adaptive decision prediction using advanced ML techniques

Explainable AI

SHAP and counterfactual explanations for transparency

Ethical Safeguards

Measuring trust, bias, and manipulation risk

A comprehensive approach to ethical, explainable neuromarketing insights

System Architecture

1

Behavioral Data Module

PsychoPy based experiments capturing gaze, mouse movement, reaction time

2

Neural Integration Module

EEG signal analysis and interpretation

3

Explainable AI Module

SHAP explanations and counterfactual scenarios

4

Federated Learning Module

Privacy-preserving decentralized model training

Design Innovation

Integration of Neural and Behavioral Data

Holistic approach combining multiple data sources

Dual-Layer Explainability

Multi-level transparency for predictions

Counterfactual Scenario Analysis

What-if insights for better understanding

Ethical Metrics

Trust index and bias detection built-in

Modular Scalable Architecture

Flexible and extensible system design

Future-Ready AI Solution

Commercialization

Target Users

- Marketing agencies
- Retail & e-commerce brands
- Consumer research institutions

Opportunities

- AI-driven consumer analytics market
- Neuromarketing research tools
- Ethical AI insight platforms

Revenue Model

- SaaS subscription
- Research licensing

Growing Market Opportunity

Neuromarketing industry projected to reach billions by 2028